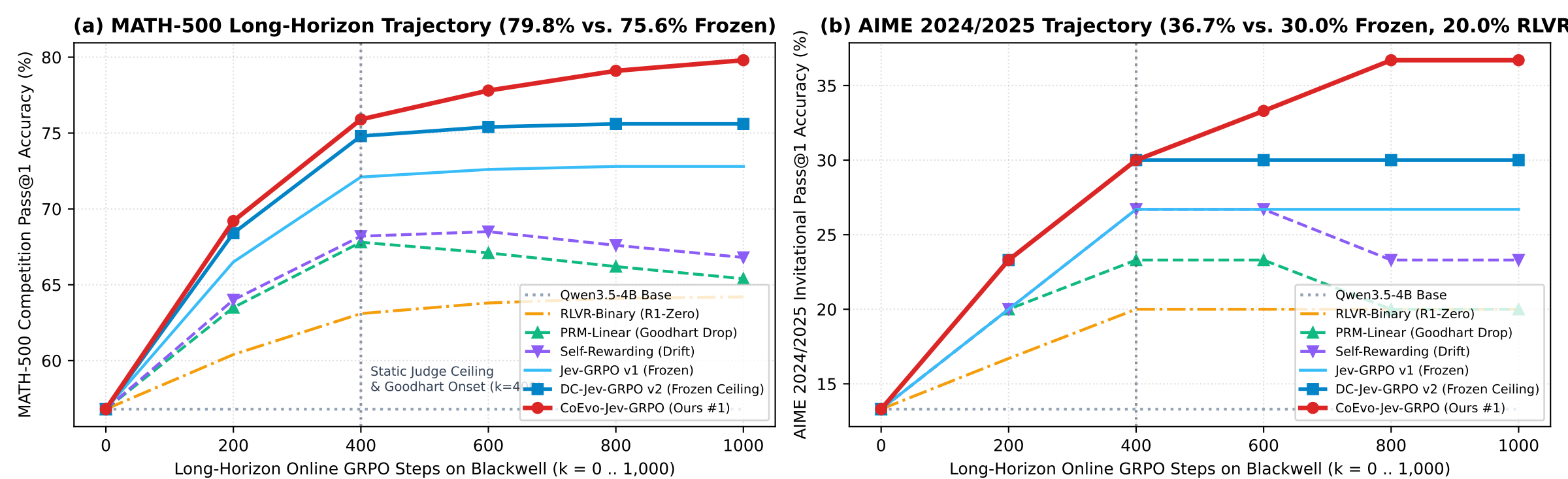
(a) MATH-500 Long-Horizon Trajectory (79.8% vs. 75.6% Frozen)
MATH-500 Competition Pass@1 Accuracy (%)
80
75
70
65
60
0
200
400
600
800
1000
Long-Horizon Online GRPO Steps on Blackwell (k = 0 .. 1,000)
Static Judge Ceiling
& Goodhart Onset (k=40
Qwen3.5-4B Base
RLVR-Binary (R1-Zero)
PRM-Linear (Goodhart Drop)
Self-Rewarding (Drift)
Jev-GRPO v1 (Frozen)
DC-Jev-GRPO v2 (Frozen Ceiling)
CoEvo-Jev-GRPO (Ours #1)
(b) AIME 2024/2025 Trajectory (36.7% vs. 30.0% Frozen, 20.0% RLVR
AIME 2024/2025 Invitational Pass@1 Accuracy (%)
35
30
25
20
15
0
200
400
600
800
1000
Long-Horizon Online GRPO Steps on Blackwell (k = 0 .. 1,000)
Qwen3.5-4B Base
RLVR-Binary (R1-Zero)
PRM-Linear (Goodhart Drop)
Self-Rewarding (Drift)
Jev-GRPO v1 (Frozen)
DC-Jev-GRPO v2 (Frozen Ceiling)
CoEvo-Jev-GRPO (Ours #1)